



Pressure-Flow experiments, packing, and modeling for scale-up of a mixed mode chromatography column for biopharmaceutical manufacturing



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ABSTRACT

To obtain consistent chromatographic behavior, it is important to develop resin packing methods in accordance with the characteristics of each resin. Resins, particularly those with a significant level of compressibility, require proper knowledge of the packing methodology to ensure scalable performance. The study demonstrates the applicability of pressure-flow modeling based on the Blake-Kozeny equation for cellulose based resins, using the MEP HyperCel (Pall) resin as a case study. This approach enabled the understanding of the appropriate bed compressibility and the determination of the minimum column diameter that can predict bed integrity during commercial manufacturing scale operation. Studies suggested that scale-dependent wall effects become negligible for column diameters exceeding 20 cm. Pressure-flow modeling produced a minimum compression recommendation of 0.206 for the MEP HyperCel resin. Columns with diameters up to 80 cm packed with this bed compression yielded incompressible beds with pressure-flow curves consistent with model predictions. Model parameter (particle diameter, viscosity, porosity) values were then varied to demonstrate how changing operating conditions influence model predictions. This analysis supported the successful troubleshooting of unexpected high pressures at the commercial manufacturing scale using MEP HyperCel resin, further supporting the applicability of this approach.

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1. Introduction

Column chromatography is used extensively in the biopharmaceutical industry to purify therapeutic proteins from complex feed streams. Development of commercial scale purification processes often employs column re-use (cycling) and/or column scale-up [1] approaches. To obtain consistent chromatographic behavior over a column's lifetime and during scale-up, it is important to develop resin packing methods in accordance with the characteristics of each resin [2,3].

Further, column operation can sometimes impact the column integrity over the course of a column lifetime if the column is deficiently packed in some manner (e.g., insufficient bed compression). Changing bed stability can lead to issues (e.g., abnormally high pressure drops across the bed, changing product pool vol-

umes, or variable product quality) that can result in flow constraints and lost throughput. Chromatography development is typically performed using scale-down models, sometimes with pre-packed columns, which have been shown to be well packed for numerous stationary phases [4]. With scale-up however, as the column diameter increases while maintaining constant bed height and superficial flow velocity, some effects at the small scale (e.g., forces exerted by the walls of the chromatography column hardware) can differ from those at larger scales. Chromatography operating conditions (e.g., bed heights, flow rates, or buffer solutions) identified at the small scale may not produce comparable product quality or process performance if the packing quality or wall effects change over the scaling-up. Upon scale-up at constant bed height, the resin compression will increase with increasing column diameter and concomitant decreases in the column aspect ratio (bed height divided by diameter). This is due to a phenomenon that as columns are scaled-up and as aspect ratios increase, the wall support to the packed bed decreases. Flow can induce increased compression of the resin, causing increased pressure drops

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across the packed bed and decreased bed stability. A further complication is that chromatography resins can exhibit varying degrees of compressibility. Resins with a significant level of compressibility may be particularly sensitive to the operating scale and packing conditions.

Several models have been developed to predict the pressure drop across packed beds at varying scales [5]. Stickel and Fotopoulos [6] developed an empirical model which correlates bed compressibility with column aspect ratio and superficial flow velocity. This model accurately predicts the critical velocity (u_{crit}) at which flow induced resin compression reaches its maximum. This compression is the critical compression (λ_{crit}), which is a resin specific parameter and is largely unaffected by column geometry. Columns packed at or above this compression behave as if they are incompressible, thus following the Blake–Kozeny equation [7,8,9] for incompressible media. Pressure-flow modeling based on the Blake–Kozeny equation can be implemented to understand bed compressibility, wall effects, and determine the minimum diameter for a representative scale-down model that can accurately predict bed integrity and performance at commercial manufacturing scale.

While this methodology was previously applied by Stickel [6] and Keener [5] to Sepharose and methacrylate-based resins having affinity, hydrophobic interaction and ion exchange functionalities, we expand upon this prior work by studying a cellulose based compressible resin with a mixed-mode ligand functionality, MEP HyperCel (Pall). MEP HyperCel is an industrially relevant mixed-mode chromatography resin [4,10–15]. While columns packed with this resin behaved consistently at small scale, prior to the application of the approach described in this paper we observed several issues during the operation of MEP HyperCel chromatography columns at the large scale. These included elevated pressures, visible bed degradation (cracks), and rising concerns about potential impact on column lifetime and/or product quality. Further investigation in-house showed pH dependent changes in particle diameter (from a mean particle diameter of 85 microns at pH ≥ 5.5 during equilibration, load and post-load washes to 89 microns at pH 3.0 during the strip). At the compression factors used to pack these columns (≤ 0.248), this resulted in an unstable packed bed and the manufacturing issues described. As these pH effects could not be avoided, modeling to determine appropriate column packing (e.g., bed compression) was performed.

The Stickel–Fotopoulos approach [6] was applied for the development of a scalable packing method for the cellulose-based mixed-mode chromatography resin MEP HyperCel. Pressure-flow modeling based on the Blake–Kozeny equation was implemented. Optimal packing compression was determined based on model predictions.

The modeling results were verified for various MEP HyperCel column packing and operating conditions at the commercial manufacturing scale. The influence of process solution viscosity, pH, resin slurry concentration, and accuracy of resin slurry delivery to the column on the recommended column packing predictions were illustrated. This was not only useful for the development of the best packing method for this resin, but was also useful to troubleshoot chromatographic behavior at the manufacturing scale. This was shown through a case study in which unexpected high-pressures were observed and the application of the modeling was used to identify the root cause.

2. Experimental

2.1. Materials

Sodium phosphate monobasic (monohydrate), sodium phosphate dibasic (heptahydrate), sodium citrate dihydrate, citric acid

monohydrate, and sodium hydroxide were purchased from Avantor (Center Valley, PA).

MEP HyperCel™ was purchased from Pall (Port Washington, NY).

2.2. Apparatus

Laboratory-scale columns (Vantage-L, Millipore, Billerica, MA) had inner diameters of 1.15, 2.2, and 4.4 cm. Pilot-scale columns (BPG, GE Lifesciences, Pittsburgh, PA) had inner diameters of 20, 30, and 40 cm. Manufacturing scale columns (Euroflow Resolute DAP, Pall Corporation Port Washington, NY) had inner diameters of 80 and 140 cm. All columns used had adjustable heights.

ÅKTA Explorer 100 FPLC systems (GE Lifesciences, Pittsburgh, PA) were used with laboratory-scale columns. PK chromatography systems (Pall Corporation, Port Washington, NY) were used with pilot and manufacturing scale columns. Pressure drops across the columns were monitored using the ÅKTA/Resolution system and/or calibrated analog pressure gauges (Pellicon 0–60 psi gage, Millipore, Billerica, MA).

2.3. Procedures

2.3.1. Packing pressure-flow curve generation

All equipment and materials were equilibrated to ambient temperature (15–25 °C) prior to use. A pressure-flow curve without the column in-line, generated for each chromatography system, was used to subtract the equipment pressure drop from the total pressure drop to enable measurement of the pressure drop across the packed bed alone. All pressure-flow profiles have the equipment pressure drop subtracted. Before packing, the resin slurry concentration was determined by one of two methods: 1) allow a sample to gravity settle overnight in a graduated conical tube or 2) centrifuge a sample for 10 min at 1600 G followed by a 30 min static hold in graduated conical tube. To begin packing, a measured volume of slurry was poured into the column. The top adaptor was inserted into the slurry and flow started. To generate a packing pressure-flow curve the procedure outlined by Stickel and Fotopoulos [6] was used: an initial flow rate generating less than 5 psig pressure drop was maintained until the bed consolidated. The flow rate was then increased incrementally, with pressure drop and bed height recorded at each increment. This was repeated until a non-linear response in pressure for an incremental change in pump speed was observed. Packing pressure-flow curves for MEP HyperCel were generated using acidic strip buffer (100 mM sodium citrate pH 3.0) mobile phase.

2.3.2. Post-pack pressure-flow curve generation

Columns were packed according to internal packing procedures to the desired bed height specifications and tested to ensure passing packed bed quality. The flow rate was then increased incrementally and the pressure drop over the packed bed recorded at each flow rate increment. The final condition (flow rate and pressure drop) did not exceed vendor recommendations to avoid further bed compression. Post-packing pressure-flow curves for MEP HyperCel were generated in both citrate buffer (100 mM sodium citrate pH 3.0) and sodium hydroxide (0.1 N) mobile phases.

2.3.3. Modeling methodology

Modeling of pressure-flow behavior in columns was performed as described by Stickel and Fotopoulos [6]. The methodology is summarized as follows:

- 1) Experimentally determine model parameters:
 - a) Packing pressure-flow curves were generated using multiple column aspect ratios as described in 2.3.1.

- b) The critical velocity (u_{crit}) was determined from each packing pressure-flow curve as the point at which the pressure-flow relationship becomes non-linear.
- c) The critical velocity times the initial bed height ($u_{crit}L_0$) was plotted against the aspect ratio (L_0/D). This yielded a linear relationship with the following correlation:

$$u_{crit}L_0 = m\left(\frac{L_0}{D}\right) + b \quad (1)$$

where the slope (m) and intercept (b) are empirical constants determined by linear regression. The empirical constant b is an indication of the resin's compressibility, while the constant m is a measure of changing wall support.

- d) Plot ($u_{crit}L_0$) against the inverse of the aspect ratio (D/L_0). This showed an asymptotic relationship as (D/L_0) increased (related to the intercept b from the plot of ($u_{crit}L_0$) against (L_0/D)). The asymptote indicated the point at which the wall effects became negligible.
- e) The critical compression of the resin (λ_{crit}) is determined by taking the average critical compression from each packing pressure-flow curve determined in 2.3.1. Critical compression was determined as:

$$\lambda_{crit} = \left(\frac{L - L_{crit}}{L_0}\right) \quad (2)$$

where L_{crit} is the bed height at u_{crit} determined in step (b).

- f) Other model parameters (d_p = effective particle diameter, ε_0 = gravity settled bed porosity, μ = viscosity of mobile phase solution, and K_0 = empirical constant for the Blake-Kozeny equation) can be obtained from the resin manufacture and literature tabulated values.
- 2) Use the model to predict packing pressure-flow profiles (confirmation of model fit)
 - a) For a given column geometry (L_0/D), calculate the critical velocity (u_{crit}) using Eq. (1) above.
 - b) Calculate bed compression using the following equation for a given linear flow velocity (u):

$$\lambda = \left(\frac{\lambda_{crit}u}{u_{crit}}\right) \quad (3)$$

- c) Calculate bed height for each compression:

$$L = L_0 - (\lambda * L_0) \quad (4)$$

- d) Calculate bed porosity for each compression:

$$\varepsilon = \left(\frac{\varepsilon_0 - \lambda}{1 - \lambda}\right) \quad (5)$$

- e) Calculate the pressure drop (ΔP) as a function of velocity (u) using values from Steps 1f, 2c, and 2d using the Blake-Kozeny:

$$\Delta P = \mu \frac{K_0 (1 - \varepsilon)^2}{d_p^2 \varepsilon^3} Lu \quad (6)$$

- f) Repeat steps 2a through 2e for increasing linear flow velocities (u) until the critical velocity is reached ($u = u_{crit}$) to generate the pressure-flow curve. The model predicted pressure-flow curves will match the experimentally generated pressure-flow curves from step 1a, and as such will be non-linear.
 - g) Repeat 2a-f for other column geometries (L_0/D) to see how pressure-flow behavior changes with column geometry. The steps can also be repeated for other mobile phase conditions as needed.
- 3) Use the model to predict post-pack pressure-flow profiles and scale-up of the packed column

- a) Pack columns to compression equal to or greater than the critical compression determined in step 1e.
- b) Post-pack pressure flow curves will be linear with intercept at zero and slope $\Delta P_{crit}/u_{crit}$ (where ΔP_{crit} was the pressure drop measured at u_{crit} from step 1 and/or 2f).

3. Results and discussion

3.1. Experimental determination of Stickel-Fotopoulos model parameters for MEP HyperCel

Packing pressure-flow curves for numerous packs at varying column aspect ratios were generated as described in Section 2.3.1 in the acidic strip buffer (when resin particle diameter is largest) (Fig. 1). For each trace, as the linear velocity increases during packing the pressure drop increases gradually until a critical velocity (u_{crit}) is reached, at which point there is a dramatic increase in pressure preventing further increases in flow rate.

Fig. 2a shows the various critical velocities (u_{crit}) determined from Fig. 1 multiplied by the initial gravity settled bed height (L_0) and plotted against the initial gravity settled bed aspect ratio (L_0/D). A linear fit of the experimental data yields the parameters for the Stickel-Fotopoulos pressure-flow model from the slope (m) and the y-intercept (b) of the line. The empirical constant b is the value of ($u_{crit} * L_0$) for an infinite diameter column, which provides a numerical indication of the compressibility of a resin for a particular buffer composition: the lower the b coefficient the higher is the compressibility of the resin. The empirical constant m provides a numerical indication of the changing wall support as a function of scale: the larger the slope m the more sensitive the resin is to changing wall support. The m and b values (1345 and 9920, respectively) for MEP HyperCel were greater than published values [6] for Sepharose 4FF and Sepharose 6 FF resins (GE Healthcare), which ranged from 400 to 1000 for m and 2000 to 5000 for the coefficient b . This indicates that the MEP HyperCel stationary phase is less compressible than the Sepharose resins, and that the changes in wall support more drastically impact the pressure-flow profile for MEP HyperCel than for the Sepharose resins.

Fig. 2b, the normalized critical velocity plotted against the inverse aspect ratio, shows an asymptote at lower values of $u_{crit} * L_0$. This means that as the diameters of the columns become larger for a fixed bed height, the wall effects become negligible.

As the linear velocity and pressure drop increase during packing, the resin bed height decreases (compression increases) until critical velocity (u_{crit}) is reached, at which point the bed compresses no further. This is the critical compression (λ_{crit}), and the resulting packed bed behaves as if incompressible, with scale-independent pressure-flow curves. Fig. 3 shows the critical compressions for the columns from Fig. 1. The maximum critical compression observed during the studies was 0.206.

The experimentally determined Stickel-Fotopoulos model parameters are summarized in Table 1.

3.2. Stickel-Fotopoulos model-predicted packing pressure-flow profiles for MEP HyperCel (confirmation of model fit)

The parameters in Table 1 were then used to model packing pressure-flow curves as described in Section 2.3.3. The model-predicted curves for columns of various diameters ranging from 1.1 cm to 180 cm in diameter are shown in Fig. 4.

The model predicted pressure-flow curves begin to overlap for columns having diameters greater than 20 cm, indicating that scale-dependent wall effects become negligible. This is consis-

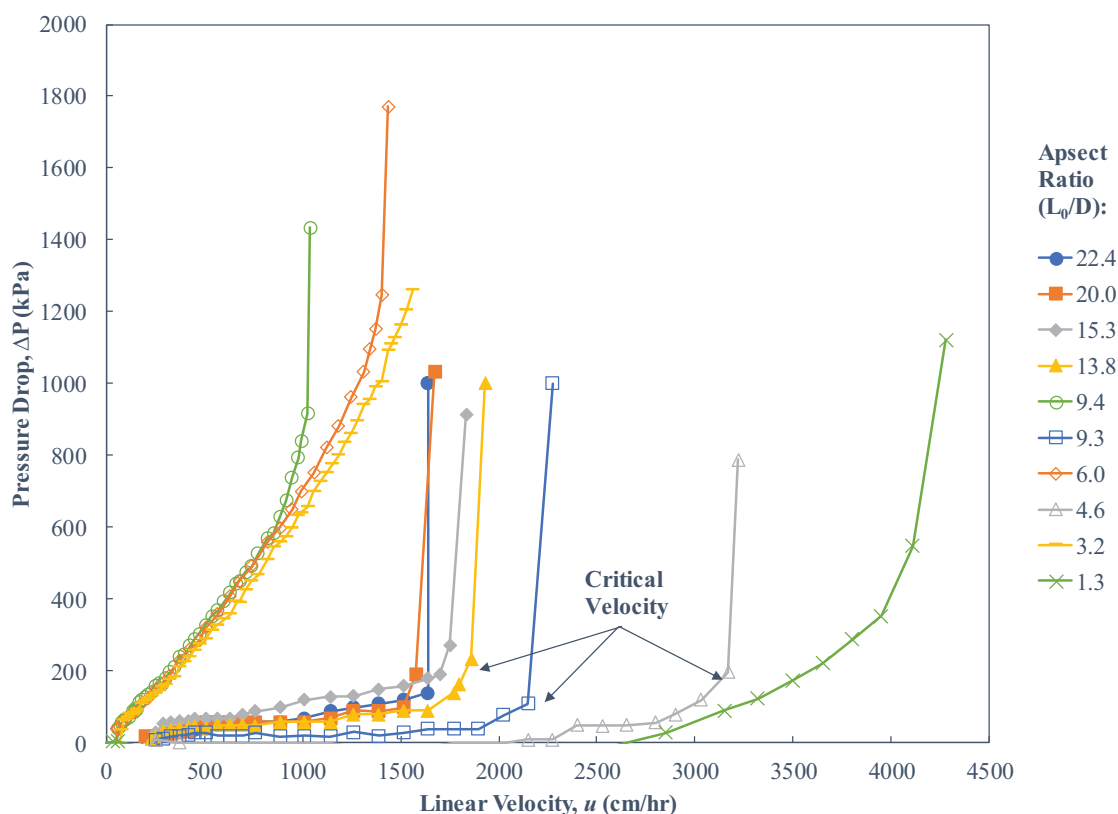


Fig. 1. Experimental pressure-flow curves for the determination the critical velocity. Pressure drop (ΔP) is plotted against the linear velocity for various aspect ratios (L_0/D).

Table 1

Stickel-Fotopoulos pressure-flow model parameters used for MEP HyperCel.

Model parameter	Value
m (cm ² /hr)	1345 ^a
b (cm ² /hr)	9920 ^a
λ_{crit}	0.206 ^b
d_p (cm)	0.00855 ^c
ε_0	0.41 ^d
μ (Pa s)	0.001037 ^e
K_0	150 ^f

^a Model parameters m and b are the slope and y-intercept, respectively, derived from a linear fit to the experimental data in Fig. 2a.

^b Maximum critical compression observed during the experiments described in Section 3.1 and Fig. 3.

^c Obtained from the resin vendor, www.pall.com.

^d Gravity settled bed porosity assumed for MEP HyperCel, based on literature [5,6] which showed the gravity settled bed porosity for 10 other commercially available resins varied between 0.38 and 0.42. An experimentally determined porosity was not obtained, as the various small molecule dye tracers injected into the column in attempt to measure porosity all irreversibly bound to the MEP HyperCel resin, preventing measurement of porosity.

^e Viscosity for the acidic strip buffer (100 mM sodium citrate pH 3.0) [16] in which the packing pressure-flow experiments described in Section 3.1 were performed.

^f Empirical constant value adopted from literature [6].

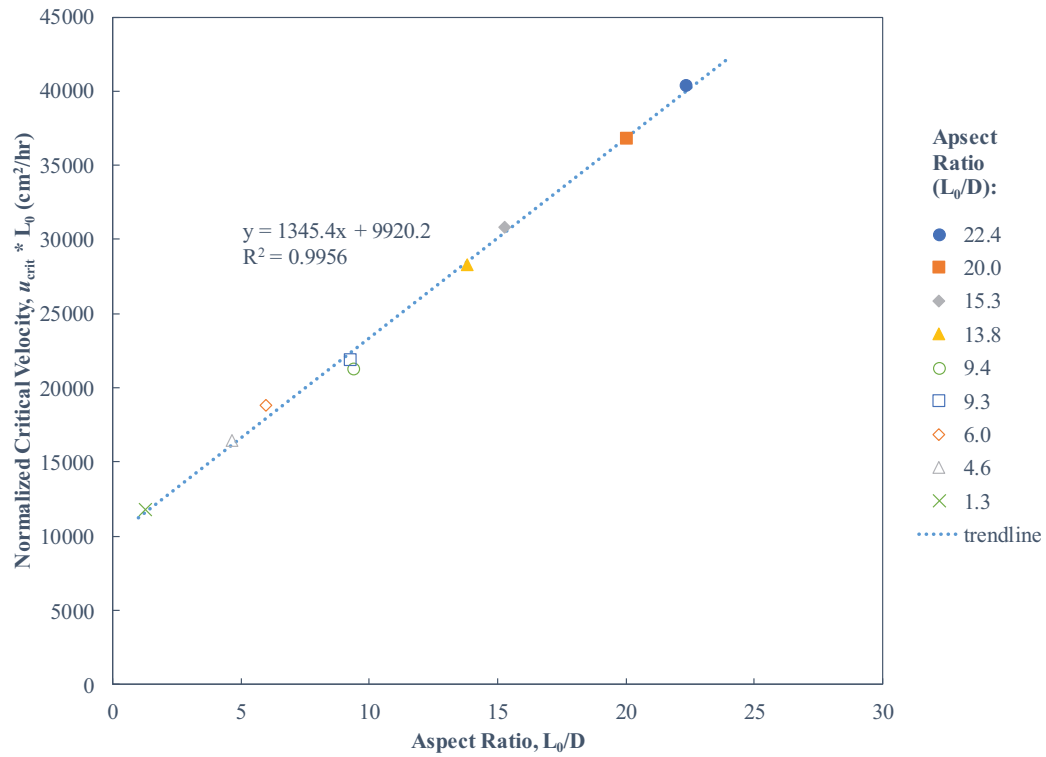
3.3. Post-pack pressure-flow profiles and scale-up of packed MEP HyperCel columns

For packed columns with stable beds, independent of column diameter scale, the column must be packed to compressions equal to or greater than the critical compression for the resin. Fig. 3 showed the maximum critical compression value observed for MEP HyperCel in acidic strip buffer to be 0.206. To confirm that this is the appropriate compression for a stable bed, two MEP HyperCel columns were packed at compression values lower and higher than the critical compression value ($\lambda = 0.130$ and $\lambda = 0.375$, respectively) and the bed stability (as measured by the number of theoretical plates, Fig. 5) and chromatographic performance (Table 2) was tested over several cycles. For the column packed at a compression of 0.130, the number of theoretical plates decreased, visible cracks formed in the column bed and changes in chromatogram shape and product volume were observed with increasing numbers of cycles of the biopharmaceutical separation. This suggested instability of the packed bed and loss of chromatographic resolution. Conversely, the column packed at a compression of 0.375 maintained its number of theoretical plates, bed integrity and chromatographic performance. These results confirmed that packing the column at or above the critical compression value yields a stable packed bed, and a compression value of 0.375 was chosen as the target compression for all future MEP HyperCel columns to be packed in the manufacturing facility.

Post-pack pressure-flow profiles were then generated for multiple columns with different diameters (1.1–80 cm, constant bed height of 19 ± 3 cm) packed with MEP HyperCel at the compression value of 0.375. Fig. 6 shows the experimentally generated pressure-flow profiles, as well as the Stickel-Fotopoulos model-predicted pressure-flow profile. The plotted pressure-flow curves

tent with the experimental data presented in Fig. 2b for columns with diameters greater than 20 cm. These results confirm the fit of the Stickel-Fotopoulos model parameters in Table 1 for MEP HyperCel.

a)



b)

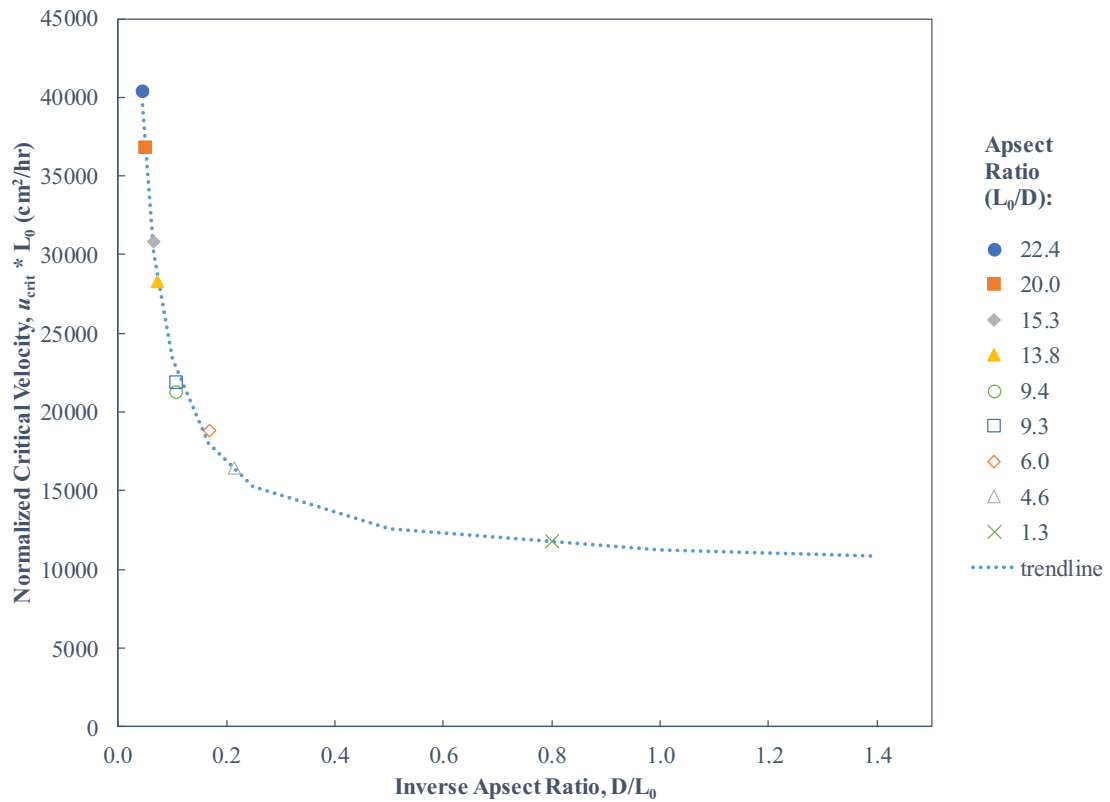


Fig. 2. a) Critical velocity times initial bed height ($u_{crit} * L_0$) plotted against the aspect ratio (initial bed height divided by the diameter, L_0/D), b) Critical velocity times initial bed height ($u_{crit} * L_0$) plotted against the inverse of the aspect ratio (diameter divided by initial bed height, D / L_0).

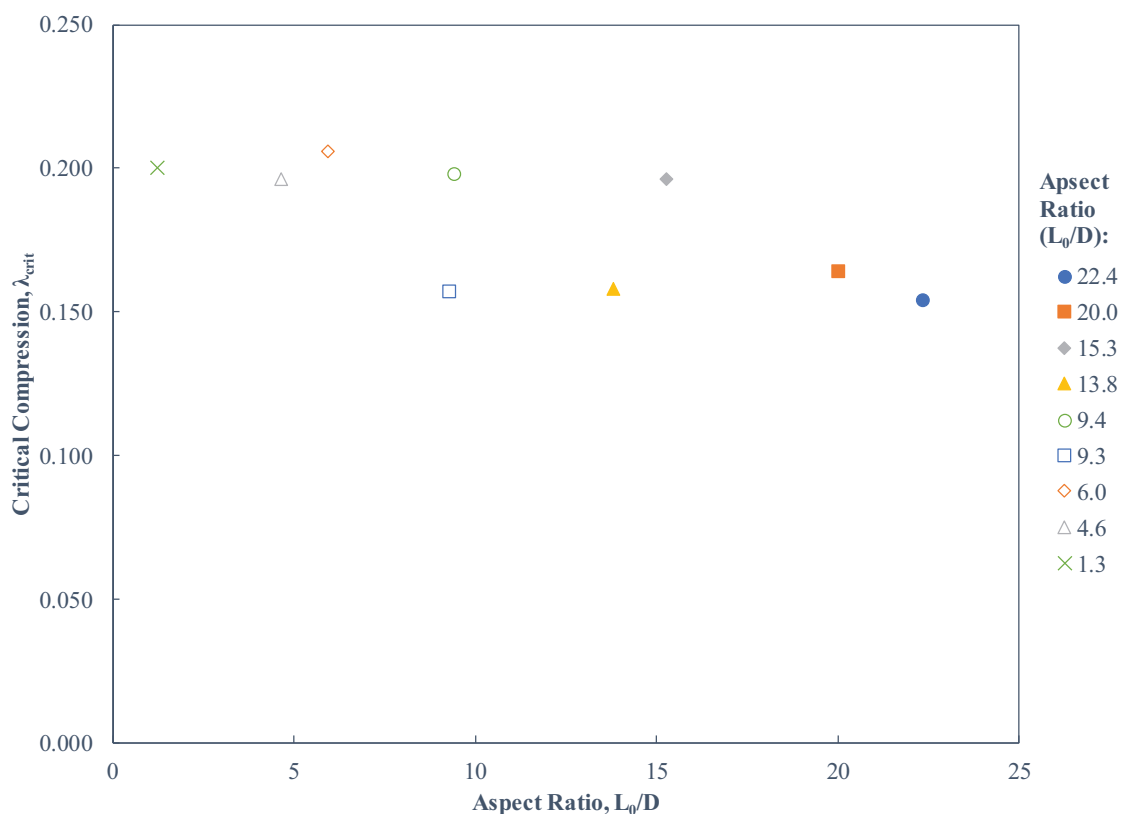


Fig. 3. Critical compressions, corresponding to the critical velocities determined in Fig. 1, plotted against aspect ratio.

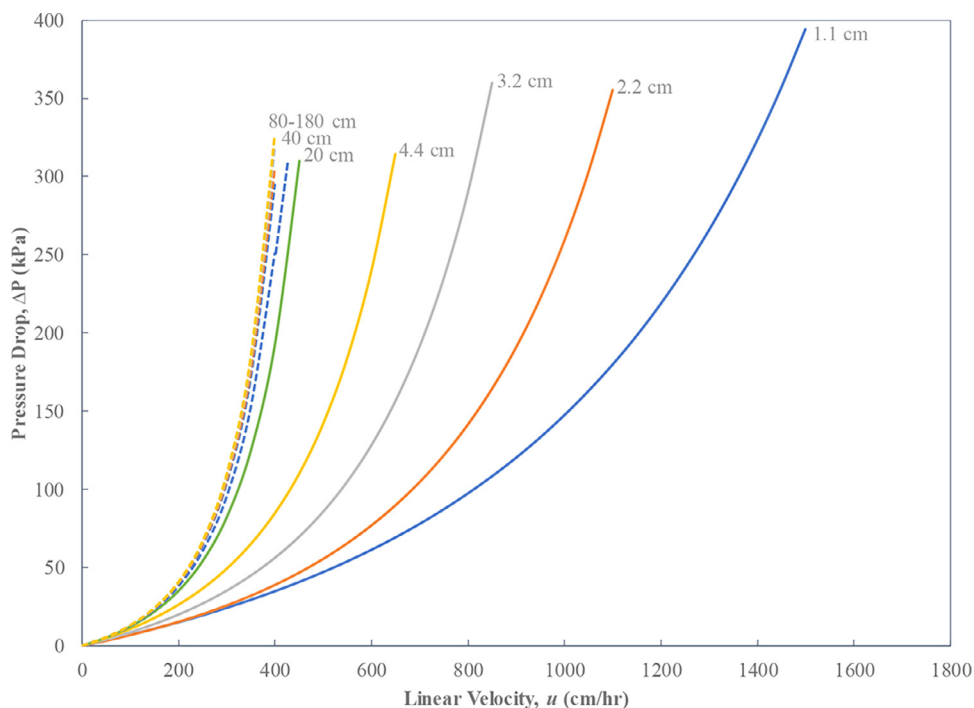


Fig. 4. Pressure-flow modeling (Stickel-Fotopoulos) for gravity settled MEP beds in varying diameter columns.

were all linear and showed good agreement with the model prediction.

3.4. Additional considerations for model application

The application of the model in the manufacturing space can predict appropriate column packing for consistent bed integrity.

The Blake-Kozeny equation used in the modeling is dependent on several parameters whose variability should be considered in applying this approach to applications in which the packed column is exposed to multiple different mobile phases as the column is cycled during processing; mobile phase viscosity, resin particle diameter, and resin porosity. This is a challenge particularly for MEP HyperCel as the resin particle diameter shrinks and swells markedly

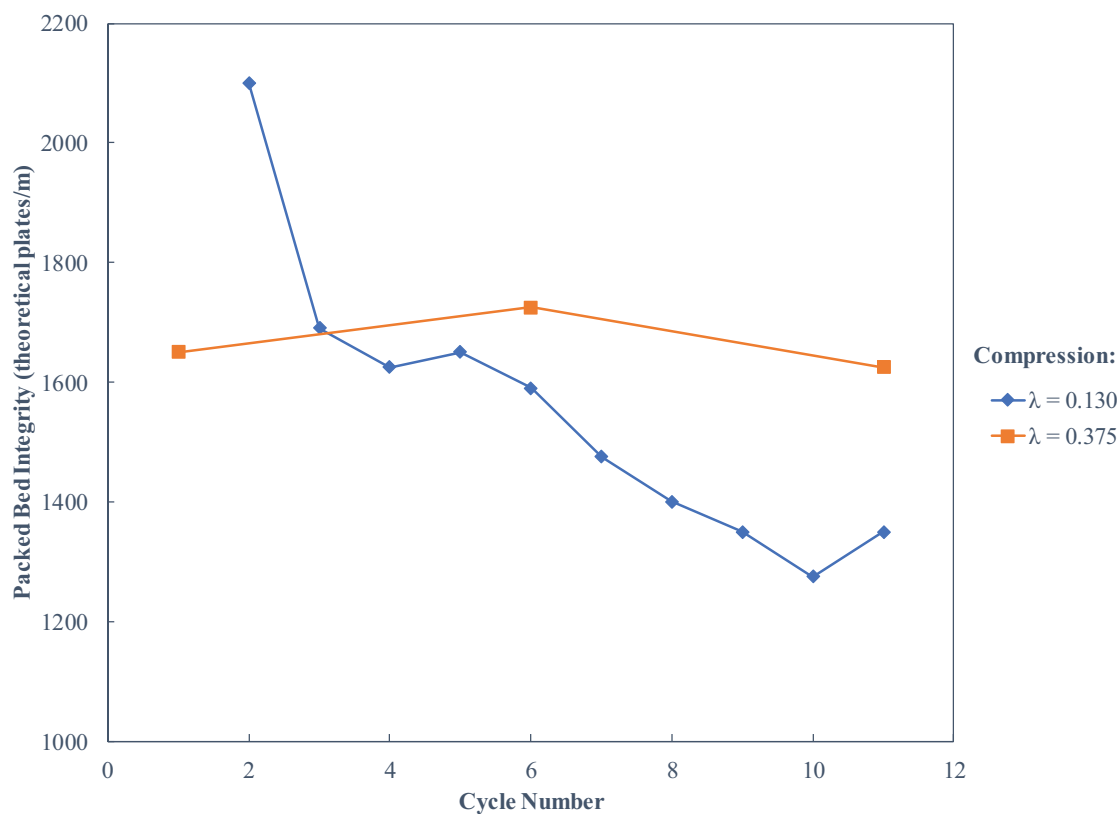


Fig. 5. Packed bed integrity, as measured by theoretical plates per meter, for MEP Hypercel columns packed with compression levels of $\lambda = 0.130$ ($\diamond$) and $\lambda = 0.375$ ($\square$).

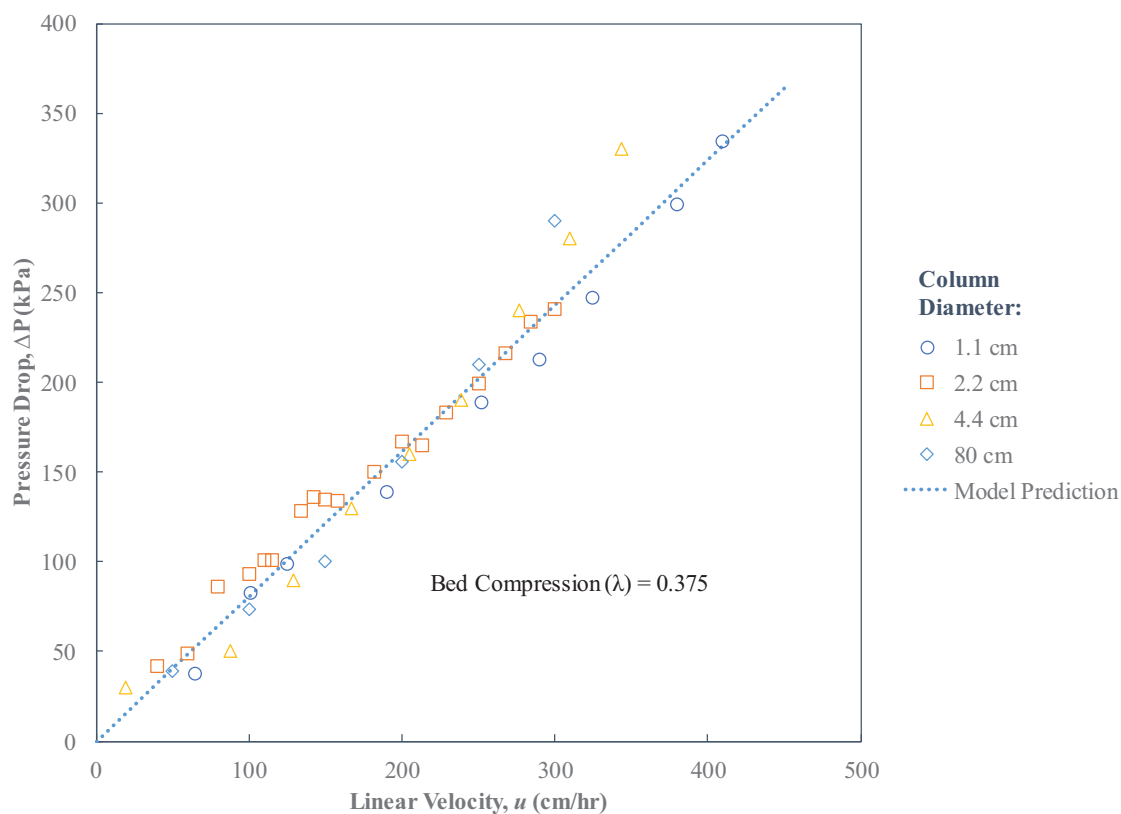


Fig. 6. Experimental post-pack pressure-flow curves for varying diameter columns packed with MEP HyperCel at a bed compression of 0.375.

Table 2

Packed bed integrity and its influence on chromatographic performance for MEP HyperCel columns packed at compression levels of $\lambda = 0.130$ (below critical compression) of $\lambda = 0.206$ and $\lambda = 0.375$ (above critical compression).

Number of product contact cycles:		1	6	11
$\lambda = 0.130$ (below critical compression)	HETP (plates/m)	2100	1600	1350
	Column visual appearance	No visual defects	No visual defects	Visible cracks
	Chromatogram shape	Reference chromatogram	Breakthrough during load, pre-peak before elution	Breakthrough during load, large pre-peak before elution, elution peak tailing
	Product volume (CV)	4.6	5.9	11.2
$\lambda = 0.375$ (above critical compression)	HCP (ppm)	Not tested		
	HETP (plates/m)	1650	1700	1650
	Column visual appearance	No visual defects	No visual defects	No visual defects
	Chromatogram shape	Reference	Consistent with reference	Consistent with reference
	Product volume (CV)	5.6	6.6	5.6
	HCP (ppm)	3	<5 ^a	<5 ^b

^a HCP measured after cycle 8.

^b HCP measured after >11 cycles.

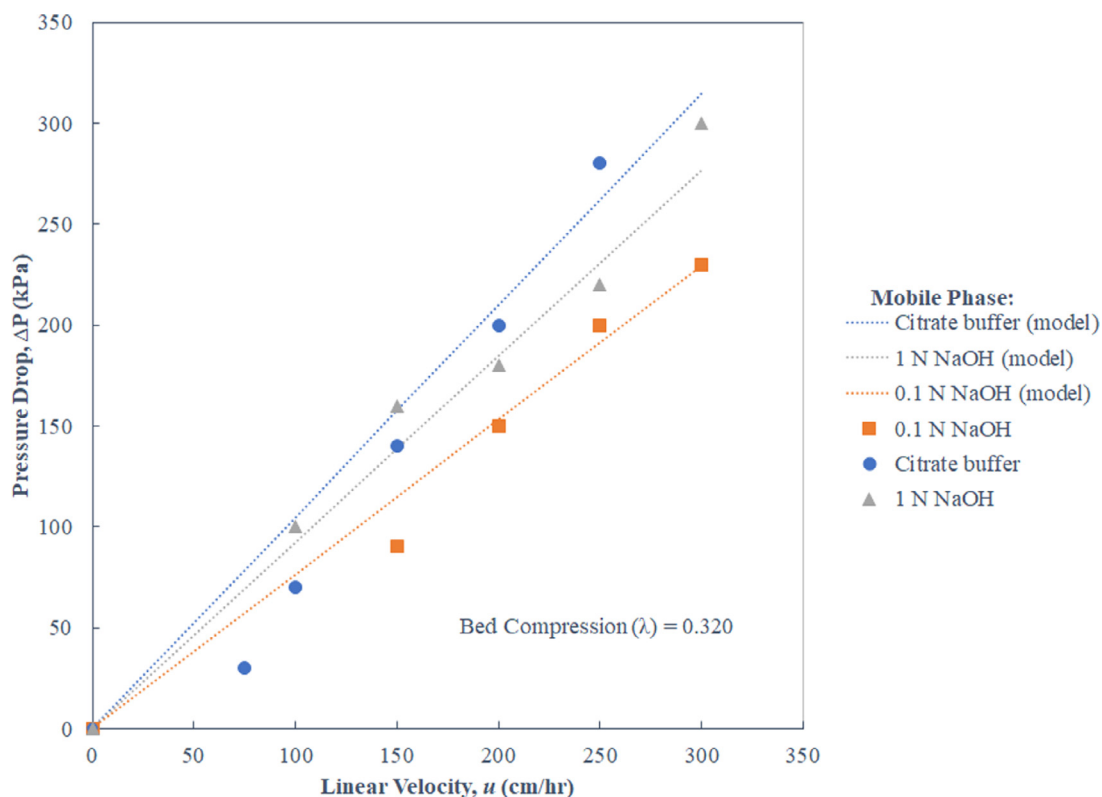


Fig. 7. Impact of mobile phase on pressure-flow behavior. Model predictions are based on Blake–Kozeny equation incorporating known values for resin bead size and mobile phase viscosity.

with shifts in mobile phase pH. Additionally, inaccuracies in measurements of the percentage of resin slurry, packing flow rate, and volume of resin slurry delivered to the column can influence the actual compression achieved and thus resin porosity. The impacts of each of these parameters on the pressure flow model are explored below, including their consideration during troubleshooting in the manufacturing setting.

3.4.1. Particle diameter

The work described in this paper thus far demonstrates the application of the model for MEP HyperCel in the acidic strip buffer. In this buffer, the MEP HyperCel particle diameter was largest (89 microns). To test the impact of resin particle diameter on behavior of MEP HyperCel columns, pressure-flow curves were generated in 0.1 N sodium hydroxide, which is the recommended packing solution for MEP HyperCel and in which the particle size was

the smallest (85 microns). Sodium hydroxide also served as a surrogate for neutral pH equilibration and wash buffers because particle size was determined to have little variability above pH 5.5. As the acid strip and 0.1 N sodium hydroxide had comparable viscosities (1.04 cP and 1.02 cP, respectively), the influence of particle diameter could be assessed independent of solution viscosity.

Fig. 7 shows the model predicted and experimental pressure-flow curves for particle diameters resulting from the two mobile phase solutions. The model predicted a 10% increase in pressure per 1 micron increase in particle diameter (calculated by changing the particle diameter value in the Blake–Kozeny equation), in agreement with the experimental data.

3.4.2. Viscosity

The influence of mobile phase viscosity was evaluated in 1 N sodium hydroxide, which had higher viscosity (1.23 cP), as com-

pared to the 0.1 N sodium hydroxide (1.02 cP) studied in the previous section [16]. As both these solutions had caustic pH, the influence of viscosity could be assessed independent of particle diameter (particle diameter was 85 microns at this pH).

Fig. 7 shows the model predictions and the experimental data for pressure-flow curves generated in the two hydroxide solutions. The model predicted a 10% increase in pressure per 0.1 cP increase in viscosity (calculated by changing the viscosity value in the Blake–Kozeny equation), in agreement with experimental data.

3.4.3. Porosity

Operational factors during column packing, such as resin slurry concentration, packing flow rate, and volume of slurry delivered, also influence model fit assumptions via the porosity (ε) term of the Blake–Kozeny equation. This was explained by Keener, et al. [5], who showed that the porosity term of the Blake–Kozeny equation is dependent on bed compression (λ) according to the following expression shown in Equation 7:

$$\varepsilon = \varepsilon_0 e^{[c(\frac{L-L_0}{L})]} = \varepsilon_0 e^{[-c(\frac{\lambda}{1-\lambda})]} \quad (7)$$

where c is a fitted parameter from the packing pressure-flow curve experiments. Adding too much resin slurry to the column or packing at too high of a flow rate results in over-compression of the bed (impacting L and λ in Eq. (7)) and a reduction in the porosity. For example, a 1% increase in amount of resin slurry added to the column results in a 1% increase in bed compression, which translates to a 2.5% decrease in porosity (calculated from Eq. (7)) and a 9% increase in pressure (calculated by changing the porosity value in the Blake–Kozeny equation).

These exercises show that no matter how good a mathematical model, a successful application needs to consider factors and variability external to the model. Multiple iterations of the model may be necessary to gain full understanding of pressure-flow behavior and ensure a stable packed bed. This investigation suggests an alternative application of the model to troubleshoot unexpected pressure-flow behavior.

3.5. Use of the model to troubleshoot unexpected chromatographic behavior

Section 3.3 demonstrated the successful application and verification of model predictions for columns between 1 cm and 80 cm in diameter. The model predictions should hold true for even larger diameter columns. However, when a 140 cm diameter column was packed in our commercial manufacturing facility with the model recommended target compression of 0.375, the post-pack pressure-flow profile generated in the acidic strip buffer showed 5-fold higher pressure than had been predicted by the model and previously demonstrated in smaller columns (Fig. 8), triggering an investigation into the root cause.

The factors discussed in Section 3.4 were scrutinized to identify the root cause of the higher than expected pressure. The correct packing buffer was selected and correctly prepared, eliminating the influence of particle diameter and solution viscosity. The pump used to deliver resin slurry and packing buffer to the column was appropriately calibrated, eliminating the influence of flow rate on porosity and bed compression. Similarly, the volume totalizer on packing system and the resin slurry tank weight totalizers were appropriately calibrated, eliminating the influence of equipment on total volume of resin delivered to the column, porosity and bed compression. Lastly, the measurement of slurry concentration was investigated. It was found that slurry concentration had been underestimated, and therefore too much resin had been added to the column. The actual bed compression was larger than intended (0.429 versus 0.375), so porosity was much lower than

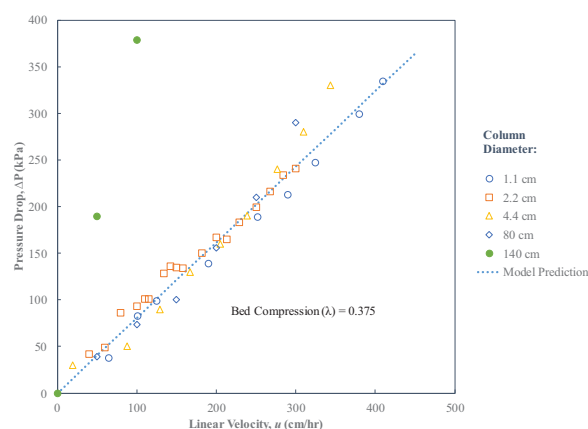


Fig. 8. The pressure-flow profile for the 140 cm diameter manufacturing-scale column (•) is accurately predicted (solid line) when the actual compression of 0.429 was used, as compared to the pressure-flow profiles and model prediction columns packed at the intended compression of 0.375 (other symbols, dashed line).

the porosity value used in the model and the resulting pressures much higher than expected (Fig. 8), consistent with the discussion in Section 3.4.3.

This investigation illustrates an alternative application of the model to troubleshoot unexpected pressure-flow behavior.

4. Conclusions

Modeling and experiments were used to gain a fundamental understanding of column performance for the mixed-mode resin MEP HyperCel.

Pressure-flow modeling was used to predict chromatographic bed compression, wall effects, and integrity at commercial manufacturing scales. Scale-dependent wall effects became negligible for columns of diameters greater than 20 cm. A bed compression greater than 0.206 yielded an incompressible resin bed resulting in scale-independent pressure-flow curves.

Further, it was demonstrated that variability in the model parameters seen during operation of the chromatography column for biopharmaceutical separations can impact the predictive capability of the model for a packed bed. Mobile phase viscosity and particle diameter (as influenced by solution pH), as well as limitations in measuring equipment for the determination of the resin slurry concentration and volume, can exacerbate the resulting bed compression and pressure drops.

Finally, the model was used to troubleshoot high pressure observations in a commercial manufacturing setting, illustrating an alternative application of the model and the model's predictive power.

Collectively this work and the prior related literature [5,6] demonstrate the robustness and consistency of this approach, and its predictive capability for many types of resins.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Jessica Prentice: Conceptualization, Methodology, Formal analysis, Investigation, Writing - original draft. **Steven T. Evans:** Conceptualization, Methodology, Formal analysis, Investigation, Writing - original draft. **David Robbins:** Conceptualization, Writing - review & editing. **Gisela Ferreira:** Conceptualization, Writing - review & editing, Supervision.

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